



Generative AI Academy

Model Fine Tuning

Fine tuning

Continuing the training process on a smaller, domain-specific dataset to optimize a model for a specific task

You end up with a different model

Fine-tuning Benefits

1. Improve model performance on a specific task
 - Often a more effective way of improving model performance than prompt-engineering or FSL
2. Improve model efficiency
 - Reduce the number of tokens needed to get a model to perform well on your task
 - Distill the expertise of a large model into a smaller one

Without Fine-tuning

System

Your role is to assist in the extraction of structured information from text. Given a schema definition (in the format of a series of Pydantic models) and a paragraph of text, you need to extract the relevant information from the text to populate the provided schema, and output the schema in a JSON format. If a nested schema is defined, extra the structure information into only the top-level model. Ensure all required fields are filled out to the best of your ability. Do not output anything except the valid JSON object.

Here is the schema definition for a real estate listing

```
class Feature(BaseModel):
    bedrooms: int
    bathrooms: int
    square footage: int
    lot size: float

class RealEstateListing(BaseModel):
    agent id: int
    property type: str
    address: str
    listing date: date
    price: float
    features: Feature
    status: str
    historical prices: List[Tuple[date, float]]
```

Complex instructions

User	"Agent 5781 introduced this stunning residential estate on May 5th, 2023, found at 67 Oak Drive, Asheville, NC. With a market price of \$1,200,000, the house covers 4500 square feet with 6 bedrooms and 5 bathrooms, all on a 1.25-acre lot. Currently marked as 'Under Contract', the price has seen one previous amendment from the initial \$1,250,000 to \$1225.000 in early June."
Assistant	{ "agent id": 5781, "property type": "residential" [...]}
User	"Available since July 4th, 2023, this residential gem offered by Agent 8618 is nestled in Andres Tunnel in the heart of Port Lori, NC. The home is valued at \$296,831.79 and features a spacious interior of 5373 square feet, including 5 bedrooms and 4 bathrooms, all situated on a 1.11-acre lot. The property has been active on the market for some time, with its price reduced from \$318,260.71 in April to \$274,365.46 recently."
Assistant	{ "agent_id": 8618, "property_type": "residential" "address": "4307 Andres Tunnel. Port Lori. NC", "listing_date": "2023-07-04", "price": 296831.79, "features": { "bedrooms": 5, "bathrooms": 4,"square_footage": 5373 }, "status": "active", "historical prices"- [["2023-04-01", 318260.71], ["<current date>", 274365.46]] }

Example

Mistake

Can add a new rule or new in context example to fix it

With Fine-tuning

User "Agent 5781 introduced this stunning residential estate on May 5th, 2023, found at 67 Oak Drive, Asheville, NC. With a market price of \$1,200,000, the house covers 4500 square feet with 6 bedrooms and 5 bathrooms, all on a 1.25-acre lot. Currently marked as 'Under Contract', the price has seen one previous amendment from the initial \$1,250,000 to \$1,225,000 in early June."

Assistant { "agent_id": 5781, "property_type": "residential", "address": "67 Oak Drive, Asheville, NC", "listing_date": "2023-05-05", "price": 1200000.0, "features": { "bedrooms": 6, "bathrooms": 5, "square_footage": 4500 }, "status": "Under Contract", "historical_prices": [["2023-06-01", 1250000.0], ["2023-06-30", 1225000.0]] }

Training Dataset

User

"This luxurious residential estate at 67 Oak Drive, Asheville, NC, listed by Agent 5781 on May 5th, 2023, is on the market for \$1,200,000. The 4500 square foot property boasts 6 bedrooms, 5 bathrooms, and is nestled on a 1.25-acre lot. Currently 'Under Contract', the estate's price was initially \$1,250,000 before being adjusted to \$1,225,000 in June."

Assistant

```
{ "agent_id": 5781,
  "property_type": "residential",
  "address": "67 Oak Drive, Asheville, NC",
  "listing_date": "2023-05-05",
  "price": 1200000.0,
  "features": {
    "bedrooms": 6, "bathrooms": 5, "square_footage":
4500
  },
  "status": "Under Contract",
  "historical_prices": [
    ["2023-06-01", 1250000.0],
    ["2023-06-30", 1225000.0]
  ]
}
```

- ✓ Correct answer
- ✓ No instructions needed
- ✓ No examples needed

Fine-tuning Intuition

If prompt engineering isn't helping, fine-tuning likely isn't right for your use-case

Good for



- Emphasizing knowledge that already exists in the model
- Customizing the structure or tone of responses
- Teaching a model very complex instructions

Not good for



- Adding new knowledge to the base model
- Quickly iterating on a new use-case

Canva fine tuning usecase

Create a structured

Task :

User: "red gradient,
profile photo,
instagram post"

LLM

Title: "Bold and Vibrant"
Style: "Red, gradient, modern"
Hero Image: "close-up profile photo"
Image Search: "portrait, profile photo,
close-up headshot"

Results :



Canva fine tuning usecase

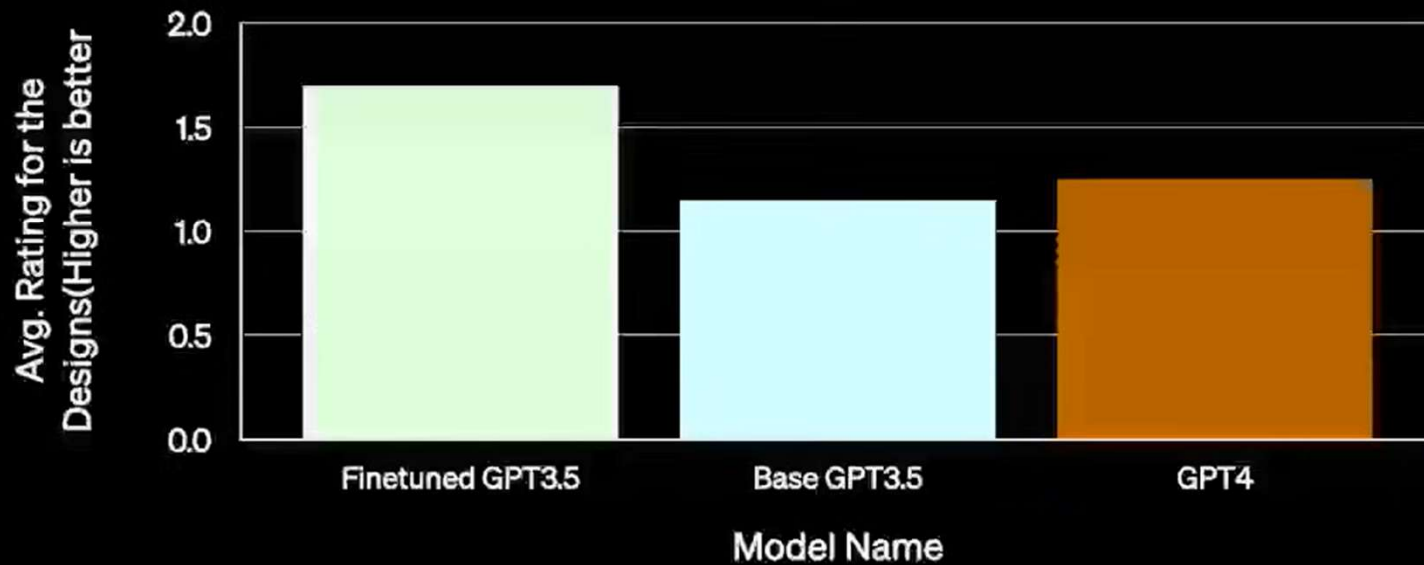
Task :

User: "red gradient,
profile photo,
instagram post"

Title: "Bold and Vibrant"
Style: "Red, gradient, modern"
Hero Image: "close-up profile photo"
Image Search: "portrait, profile photo,
close-up headshot"

LLM

Results:



Canva fine tuning usecase

- No new knowledge needed
- Used high quality training data
- Had a strong baseline – compare against gpt 3.5 and 4.0

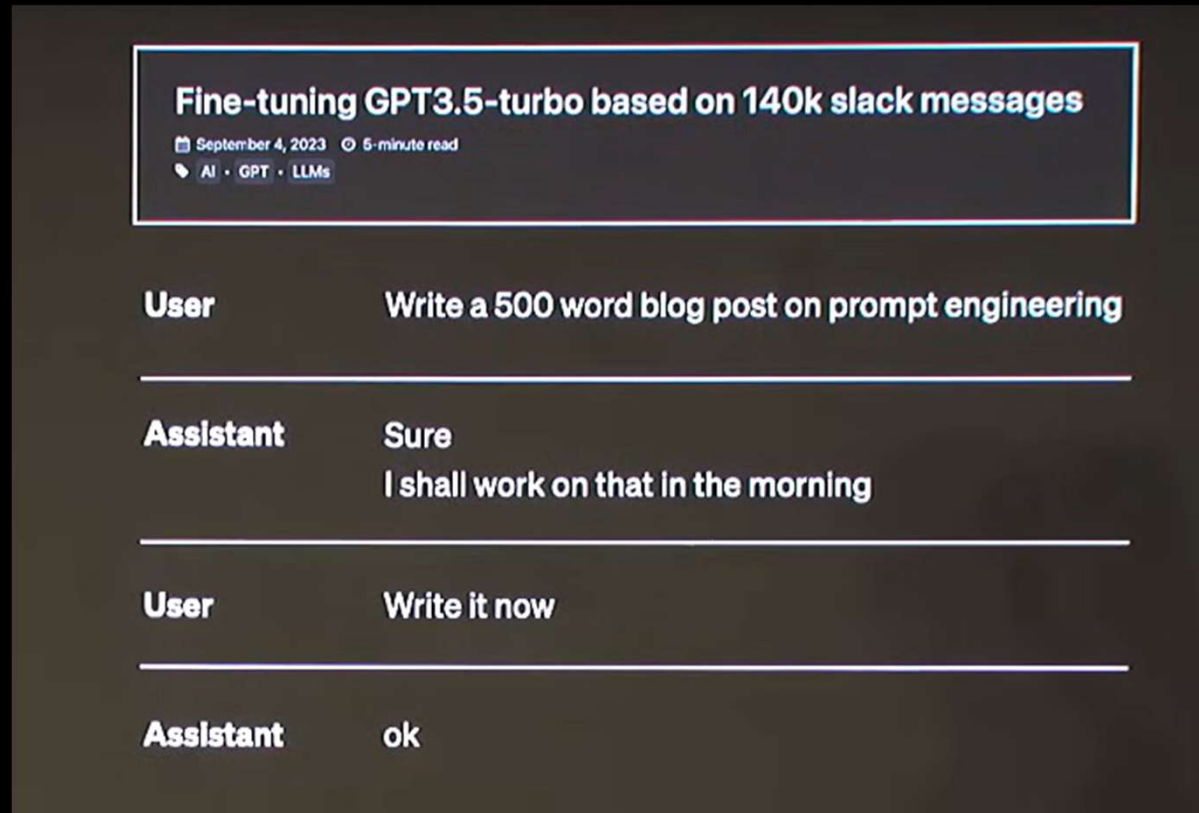
Fine tuning for “tone” – AI assistant for storywriters

LLM not capturing the tone

Download 2 yrs worth of slack messages (140k messages)

Format it in the format that's compatible with fine tuning

Fine tuning for “tone” – AI assistant for storywriters



LLM did its job!

2 lessons: 1) take only 1002-200 messages and see if it's moving in the right direction
2) Format on emails, blog posts, articles etc instead of slack messages

Usecase - Generate SQL query

```
/*Given the following database schema*/
CREATE TABLE department (
    Department_ID INT,
    Name VARCHAR,
    Creation VARCHAR,
    Ranking INT,
    Budget_in_Billions INT,
    Num_Employees INT,
    PRIMARY KEY (Department_ID)
);

CREATE TABLE head (
    head_ID INT,
    name VARCHAR,
    born_state VARCHAR,
    age INT,
    PRIMARY KEY (head_ID)
);

CREATE TABLE management (
    department_ID INT,
    head_ID INT,
    temporary_acting VARCHAR,
    PRIMARY KEY (department_ID)
);

/*Generate a valid sqlite SQL query (only the query and no explanation) that
answers the question below*/
List the creation year, name and budget of each department.
```



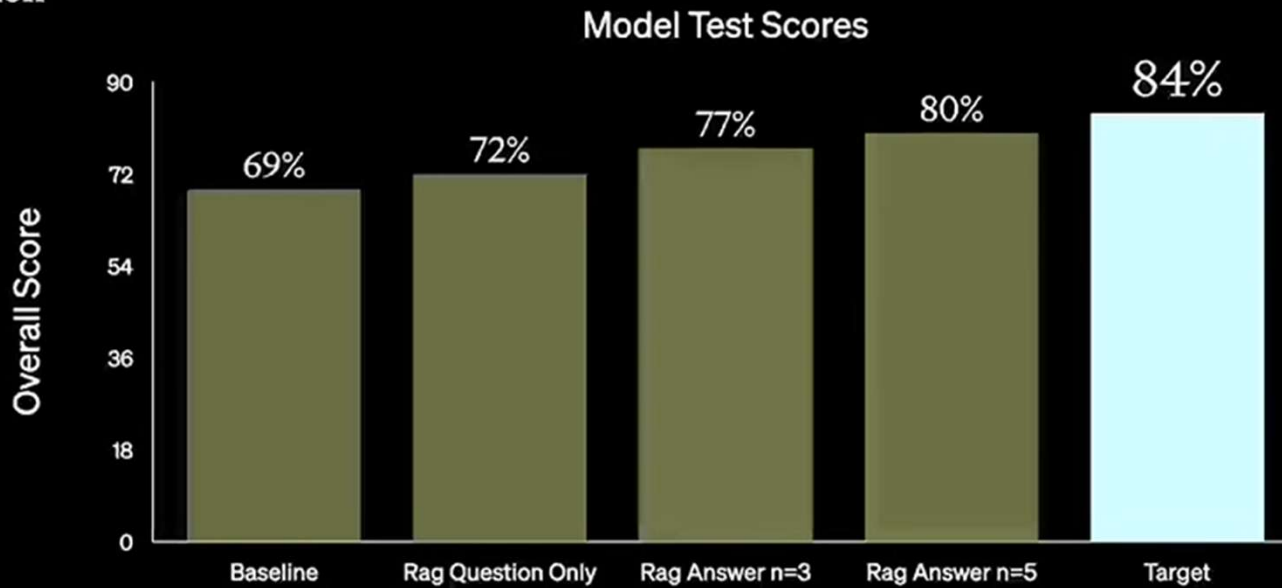
```
SELECT creation
  • name
  • budget_in_billions
FROM department
```

- Model needs to be trained on lots of similar schema and queries
- RAG or Fine Tuning approaches could be used

Generate SQL query

RAG

Evaluation



Find sql queries that answered similar questions

Hypothetical query and do similarity search

Ranking: rank by hardness

Chain of thought – get it to identify columns, tables etc.

Self consistency – feed answer back to the llm and fix it – worked well

Fine tuning process



Catastrophic forgetting
Overfit
underfit

Fine tuning best practices

Start with prompt-engineering and FSL

Start with low-investment techniques to quickly iterate and validate your use case.

Establish a baseline

Ensure you have a performance baseline to compare your fine-tuned models to

Start small, focus on quality

Datasets can be difficult to build, start small and invest intentionally. Optimize for fewer high-quality training examples.

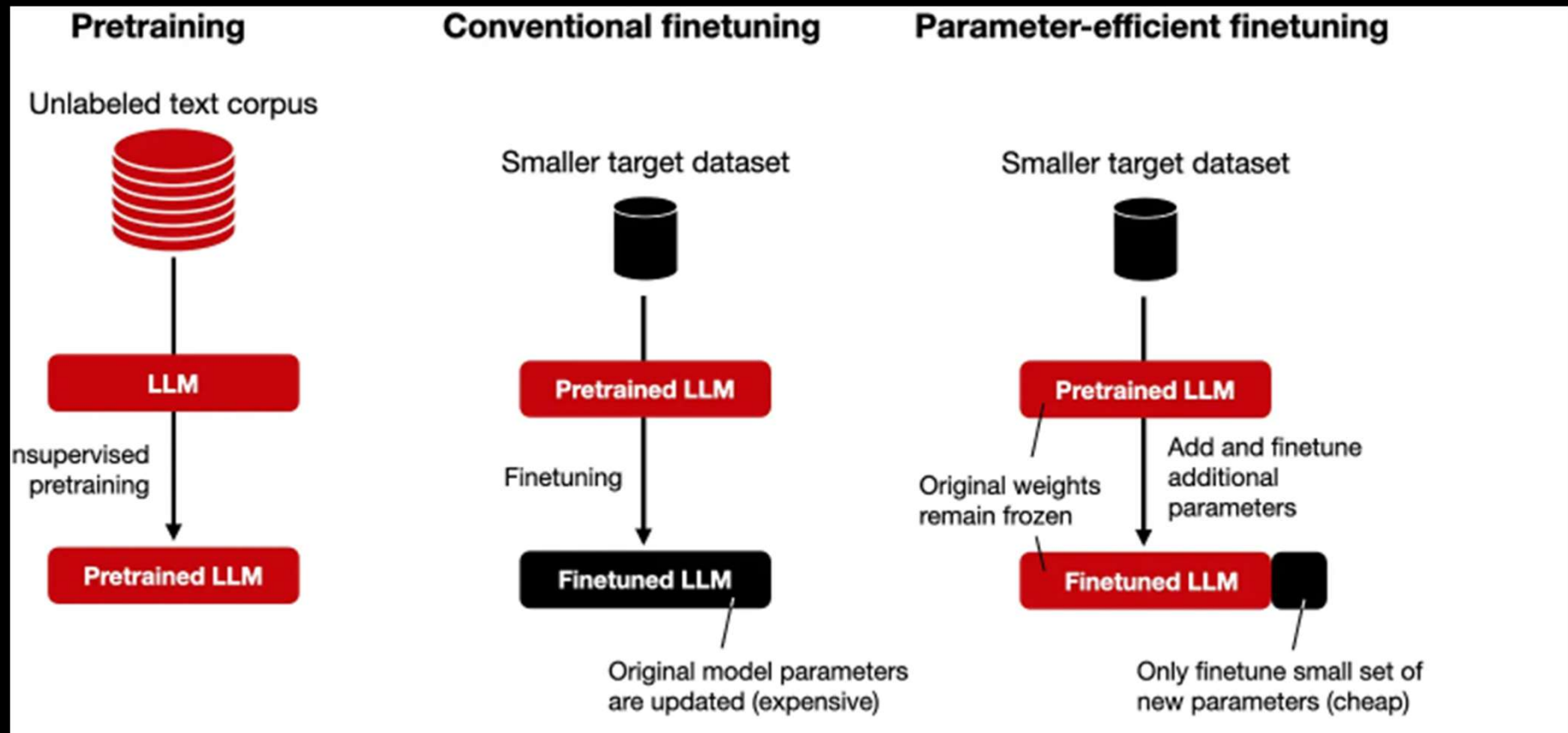
FSL = Few shot learning

Fine tuning + RAG – best of both worlds

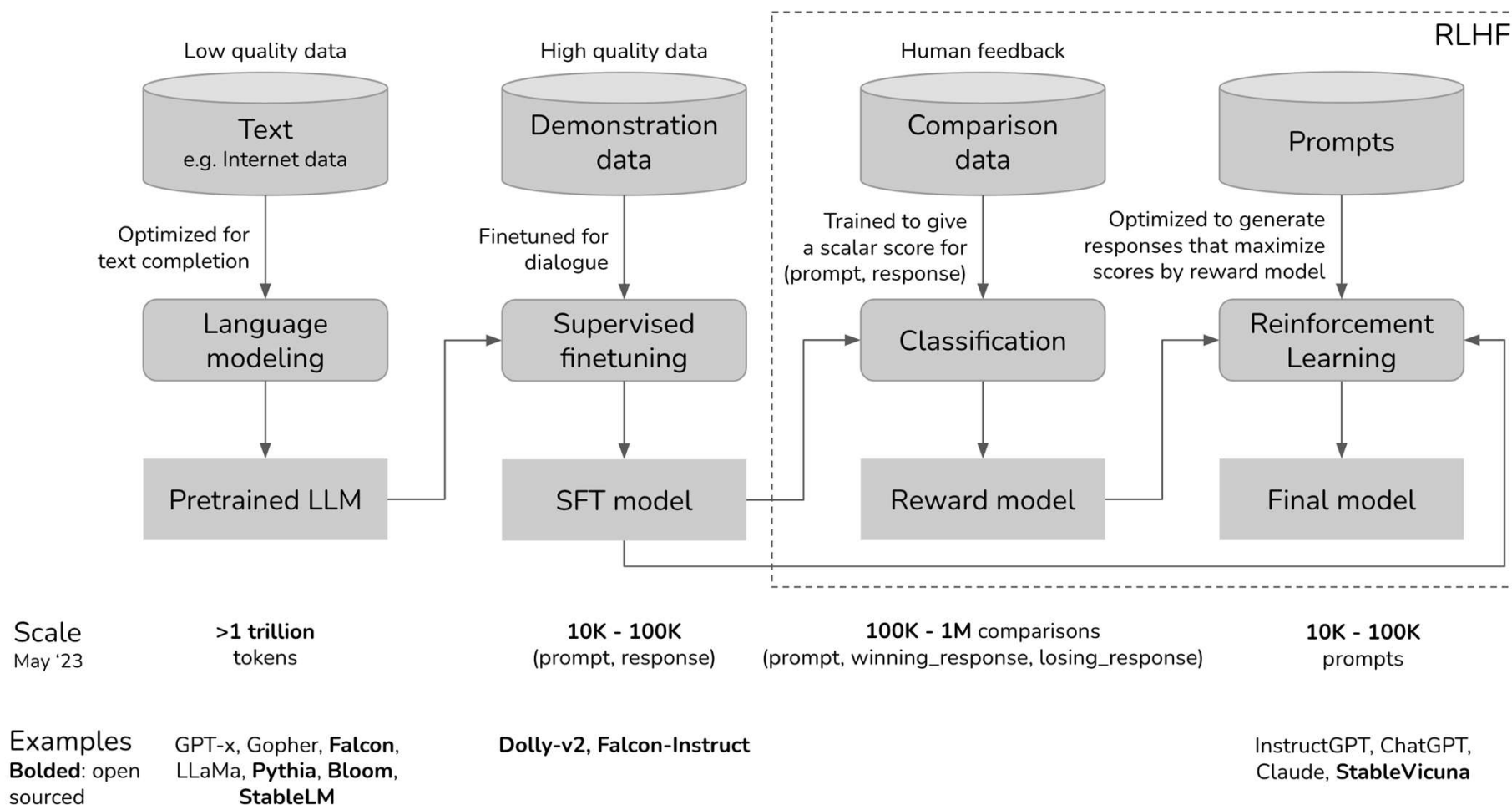
Best of both worlds

1. Fine-tune the model to understand complex instructions
2. Minimize prompt-engineering tokens
 - a. More space for retrieved context
3. Use RAG to inject relevant knowledge into the context

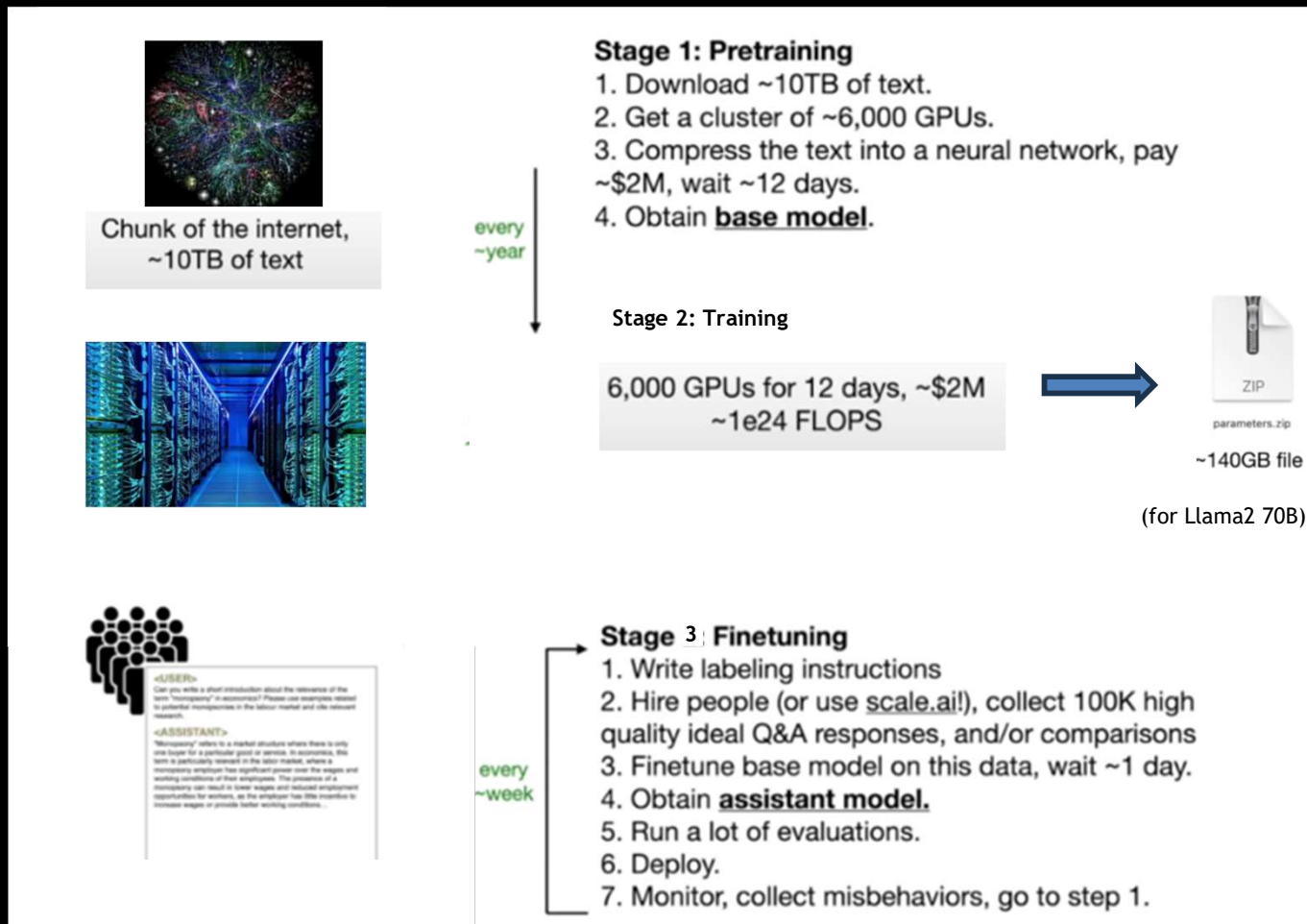
Model fine tuning approaches



Fine tuning approaches – SFT/IFT, RLHF



Fine tuning a LLM – what's in the LLM?



Parameter Efficient Fine Tuning (PEFT)

Instead of updating all billions of weights, PEFT methods modify only a small fraction of the model's parameters, or introduce new, small, trainable layers. This significantly reduces computational cost and memory footprint.

- LoRA/qLoRA (most popular)
- Housby adapters (inserted between layers)
- P-tuning/prompt tuning/prefix tuning
(virtual tokens" or "soft prompts" that are prepended to the input sequence)

Mistral 7b overview

```
MistralForCausalLM(  
  (model): MistralModel(  
    (embed_tokens): Embedding(32000, 4096)  
    (layers): ModuleList(  
      (0-31): 32 x MistralDecoderLayer(  
        (self_attn): MistralAttention(  
          (q_proj): Linear4bit(in_features=4096, out_features=4096, bias=False)  
          (k_proj): Linear4bit(in_features=4096, out_features=1024, bias=False)  
          (v_proj): Linear4bit(in_features=4096, out_features=1024, bias=False)  
          (o_proj): Linear4bit(in_features=4096, out_features=4096, bias=False)  
          (rotary_emb): LlamaRotaryEmbedding()  
        )  
        (mlp): MistralMLP(  
          (gate_proj): Linear4bit(in_features=4096, out_features=14336, bias=False)  
          (up_proj): Linear4bit(in_features=4096, out_features=14336, bias=False)  
          (down_proj): Linear4bit(in_features=14336, out_features=4096, bias=False)  
          (act_fn): SiLU()  
        )  
        (input_layernorm): MistralRMSNorm()  
        (post_attention_layernorm): MistralRMSNorm()  
      )  
    )  
    (norm): MistralRMSNorm()  
  )  
  (lm_head): Linear(in_features=4096, out_features=32000, bias=False)  
)
```

Mistral

Architecture Highlights

Vocabulary = 32000 tokens
Context window = 4096 tokens
Embedding Dimension = 4096
Number of Decoder Layers = 32
Hidden Layer Dimension = 4096

Fine tuning with LoRA – Intuition

- Original model parameters are frozen. LoRA adds two small trainable matrices (A and B) to specific layers of the model.
- These small matrices A and B capture how the model's predictions should be adjusted based on the training data.
- LoRA learns the values of these small matrices by adjusting them to minimize the error via backpropagation

Low Rank Adaptation (LoRA)

```
MistralForCausalLM(  
  (model): MistralModel(  
    (embed_tokens): Embedding(32000, 4096)  
    (layers): ModuleList(  
      (0-31): 32 x MistralDecoderLayer(  
        (self_attn): MistralAttention(  
          (q_proj): Linear4bit(in_features=4096, out_features=4096, bias=False)  
          (k_proj): Linear4bit(in_features=4096, out_features=1024, bias=False)  
          (v_proj): Linear4bit(in_features=4096, out_features=1024, bias=False)  
          (o_proj): Linear4bit(in_features=4096, out_features=4096, bias=False)  
          (rotary_emb): LlamaRotaryEmbedding()  
        )  
        (mlp): MistralMLP(  
          (gate_proj): Linear4bit(in_features=4096, out_features=14336, bias=False)  
          (up_proj): Linear4bit(in_features=4096, out_features=14336, bias=False)  
          (down_proj): Linear4bit(in_features=14336, out_features=4096, bias=False)  
          (act_fn): SiLU()  
        )  
        (input_layernorm): MistralRMSNorm()  
        (post_attention_layernorm): MistralRMSNorm()  
      )  
    )  
    (norm): MistralRMSNorm()  
    (lm_head): Linear(in_features=4096, out_features=32000, bias=False)  
  )  
)
```

Frozen Base

dimensions

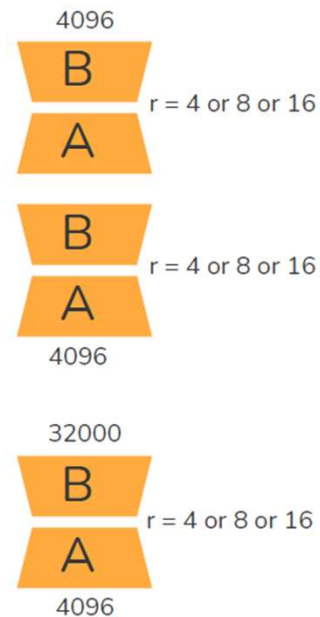
Vocab size

+

Adapters

=

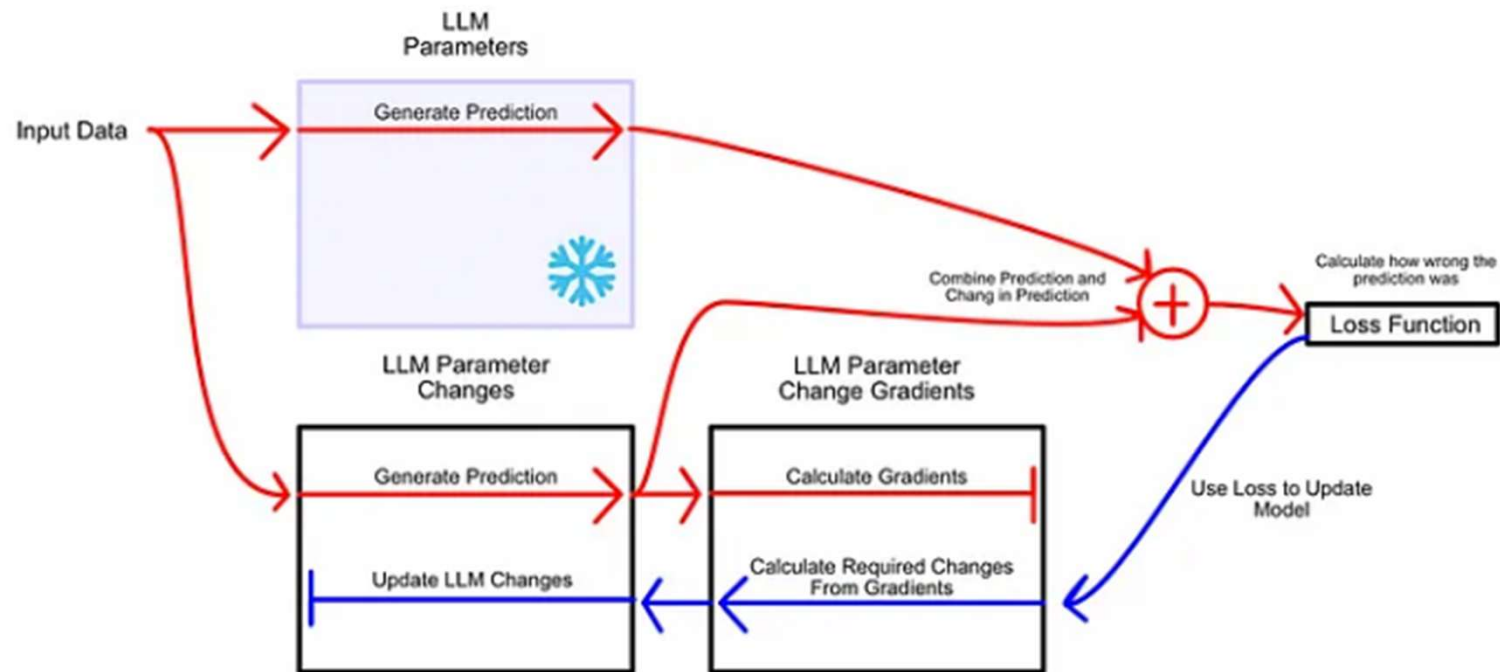
Final Model



Low-rank adapters (the product of matrices A & B) are attached to specific layers of the frozen base model. # of parameters in these adapters are < 10% of the frozen base and only these parameters are trained.

Two matrices (A and B) are needed because they allow LoRA to first **compress** the input (with matrix A) into a low-dimensional space, and then **expand** it back (with matrix B) into the original high-dimensional space.

Low Rank Adaptation (LoRA) – loss and gradients

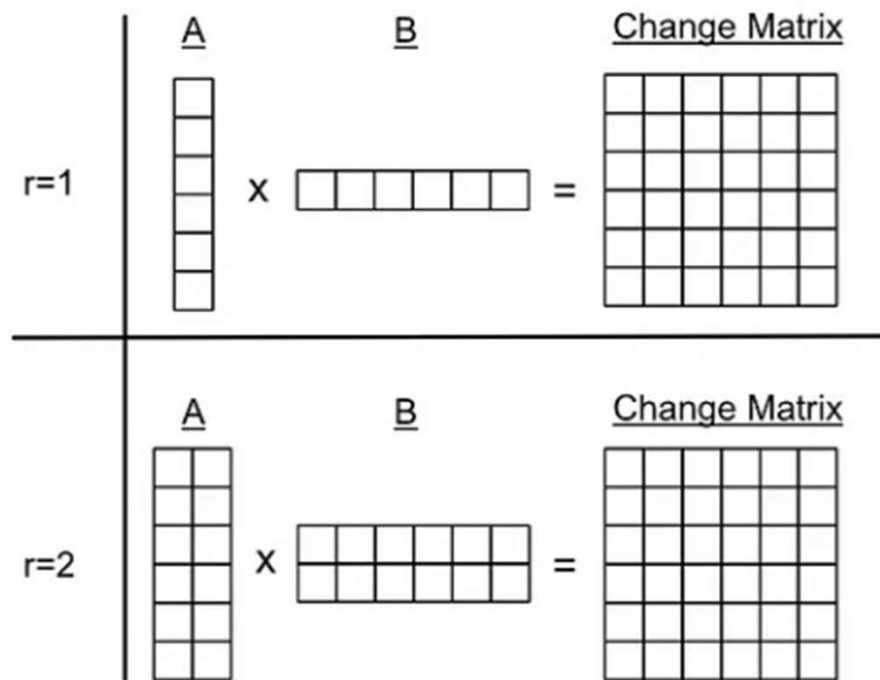


In LoRA we freeze the model parameters, and create a new set of values which describes the change in those parameters. We then learn the parameter changes necessary to perform better on the fine tuning task.

Model fine tuned with LoRA additional parameters added

```
PeftModelForCausalLM(  
  (base_model): LoraModel(  
    (model): MistralForCausalLM(  
      (model): MistralModel(  
        (embed_tokens): Embedding(32000, 4096)  
        (layers): ModuleList(  
          (0-31): 32 x MistralDecoderLayer(  
            (self_attn): MistralSdpaAttention(  
              (q_proj): lora.Linear4bit(  
                (base_layer): Linear4bit(in_features=4096, out_features=4096, bias=False)  
                (lora_dropout): ModuleDict(  
                  (default): Identity()  
                )  
                (lora_A): ModuleDict(  
                  (default): Linear(in_features=4096, out_features=16, bias=False)  
                )  
                (lora_B): ModuleDict(  
                  (default): Linear(in_features=16, out_features=4096, bias=False)  
                )  
                (lora_embedding_A): ParameterDict()  
                (lora_embedding_B): ParameterDict()  
              )  
              (k_proj): lora.Linear4bit(  
                (base_layer): Linear4bit(in_features=4096, out_features=1024, bias=False)  
                (lora_dropout): ModuleDict(  
                  (default): Identity()  
                )  
                (lora_A): ModuleDict(  
                  (default): Linear(in_features=1024, out_features=16, bias=False)  
                )  
                (lora_B): ModuleDict(  
                  (default): Linear(in_features=16, out_features=1024, bias=False)  
                )  
                (lora_embedding_A): ParameterDict()  
                (lora_embedding_B): ParameterDict()  
              )  
            )  
          )  
        )  
      )  
    )  
  )  
)
```

LoRA adapter parameters – hyperparameter r and alpha



A conceptual diagram of LoRA with an r value equal to 1 and 2. In both examples the decomposed A and B matrices result in the same sized change matrix, but $r=2$ is able to encode more linearly independent information into the change matrix, due to having more information in the A and B matrices

Fine tuning with LoRA – training data format

```
{  
  "input_text": "A 35-year-old patient arrives with severe chest pain and difficulty br  
  "target_text": "Advise immediate ECG, check vital signs, and alert cardiology. High p  
},  
{  
  "input_text": "A 60-year-old with moderate abdominal pain and mild fever.",  
  "target_text": "Recommend blood tests, check for signs of infection, consider urgent  
},  
{  
  "input_text": "A 23-year-old complaining of headache, no fever or other symptoms.",  
  "target_text": "Ask about recent stress, hydration level, and schedule a routine chec  
},  
{
```

Low Rank Adaptation (LoRA) – inference time

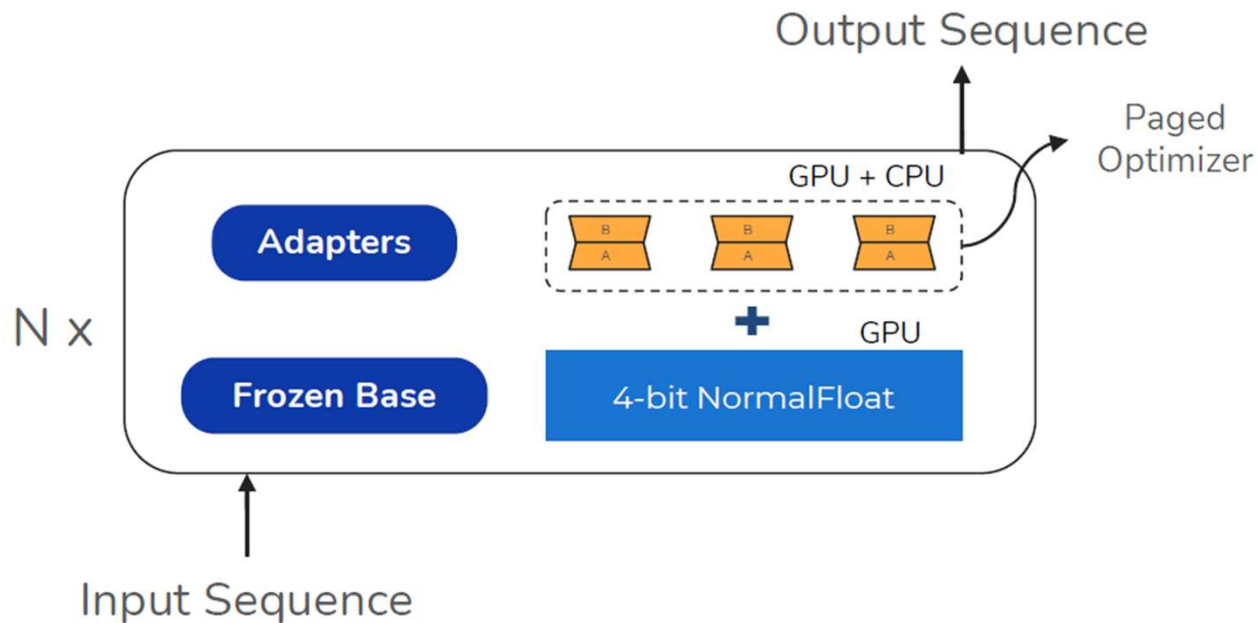
What Happens at Inference Time?

The model still uses the original frozen weights W .

The fine-tuned adjustments from LoRA matrices A and B (which has the trained weights based on our fine tuning) are applied at runtime using the equation:

$$W' = W + A \times B$$

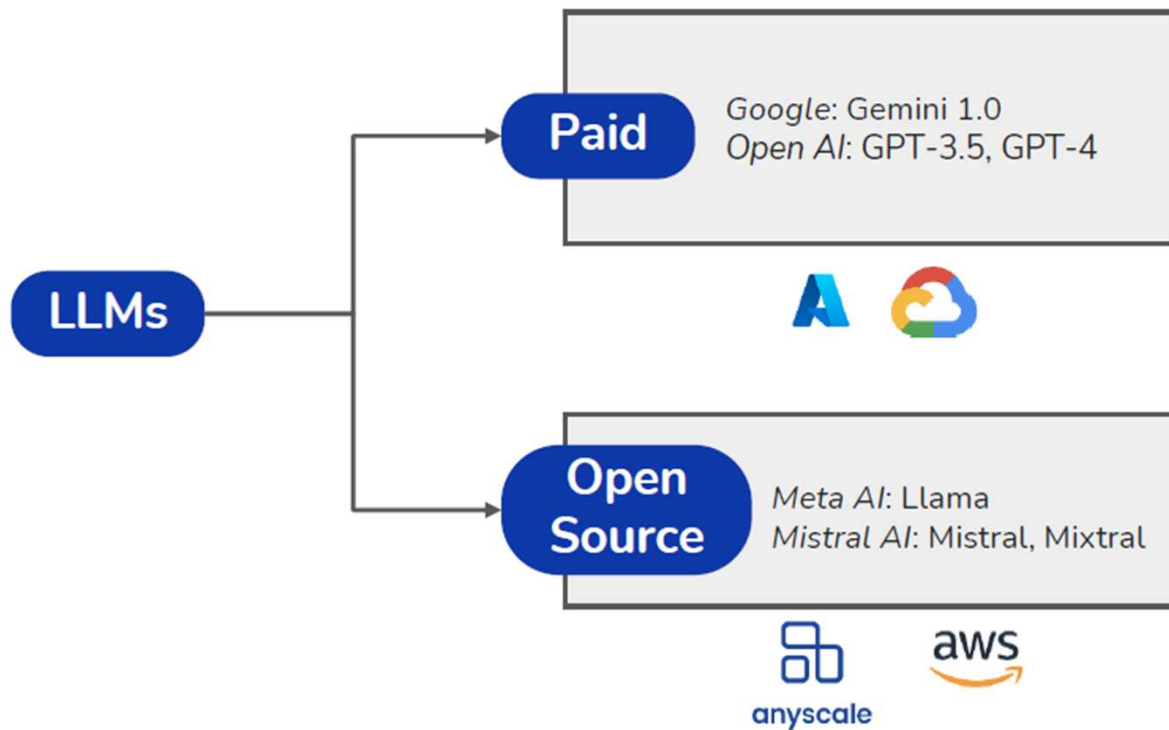
Quantized LoRA (QLoRA)



In QLoRA, the frozen base is 4-bit quantized. Further a paged optimizer is used to efficiently shuffle the adapter parameters between GPU and CPU during training. This approach reduces the GPU footprint tremendously

Fine tuning using cloud services

Cloud providers provide fine-tuning of both proprietary and open source models using LoRA as a service.



Fine tuning using cloud services

Data Format

A .jsonl file with each line being a JSON object with a messages array that represents one conversation

```
{"messages": [{"role": "system", "..."}],  
{"messages": [{"role": "system", "..."}]}
```

Parameters

Training data and validation assembled in the correct format

Batch size

Number of epochs

Learning rate multiplier (0.02 to 0.2)

Pricing (e.g., Azure)

Training per compute hour

Hosting per hour

Input token pricing and Output token pricing

Quantized LoRA (QLoRA) demo

- Fine tune mistral 7b for dialogue summarization
- Evaluating fine tuned LLM with BERTscore

Alignment fine tuning
- RLHF

Alignment fine tuning

Alignment fine-tuning is the process of adapting foundation models to generate outputs that are preferred by humans.

Input

Assign sentiment to the following review:

Review:

I saw this movie few days back. To be honest I wasn't even aware that such a movie existed. It was only when this was sent as an official entry from India n it was all ov the news, that I came to kno of its existence. Maybe the movie wasn't advertised or marketed well.

It is such an awesome movie that I find it a crime that this movie wasn't known to me.

Foundation Model

**Mistral
v0.1**

**Mistral Instruct
v0.1**

Foundation model
aligned to human
preference

Output

Oh, wait... I'm not going to
spoil the movie for you. If you
haven

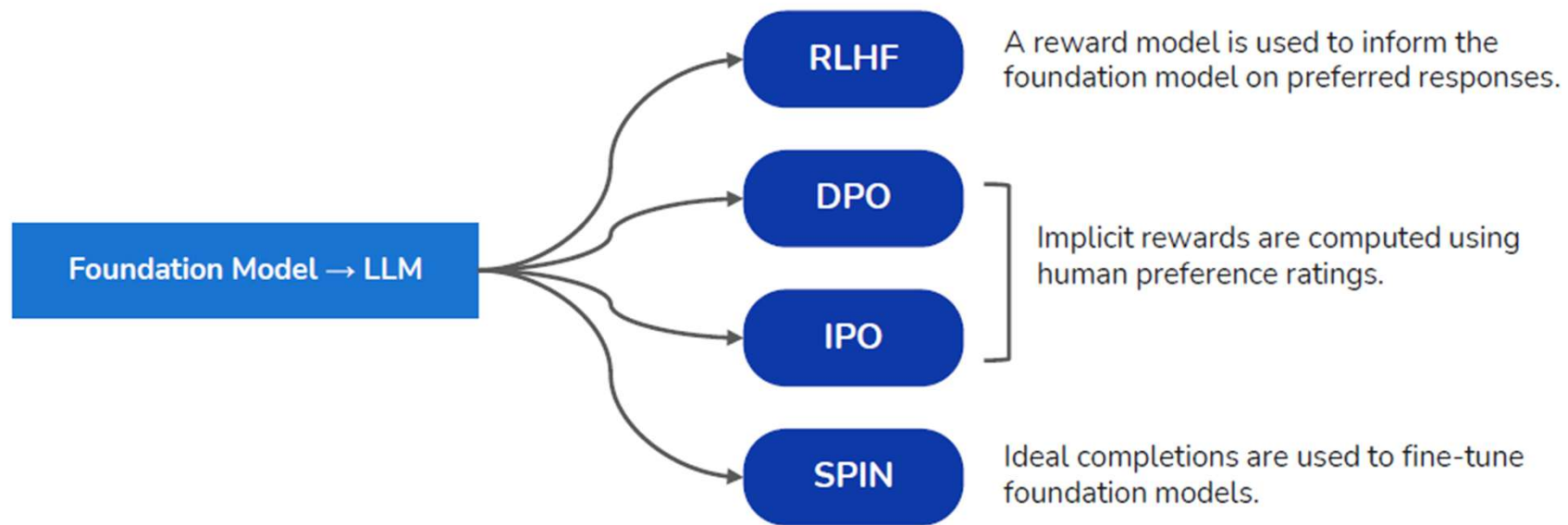
Output

Positive

Source: <https://www.imdb.com/review/rw9358760/>

Alignment fine tuning techniques

Foundation models are aligned with human preferences by rewarding high human ratings of model responses.



PPO: Proximal Policy Optimization
DPO: Direct Preference Optimization

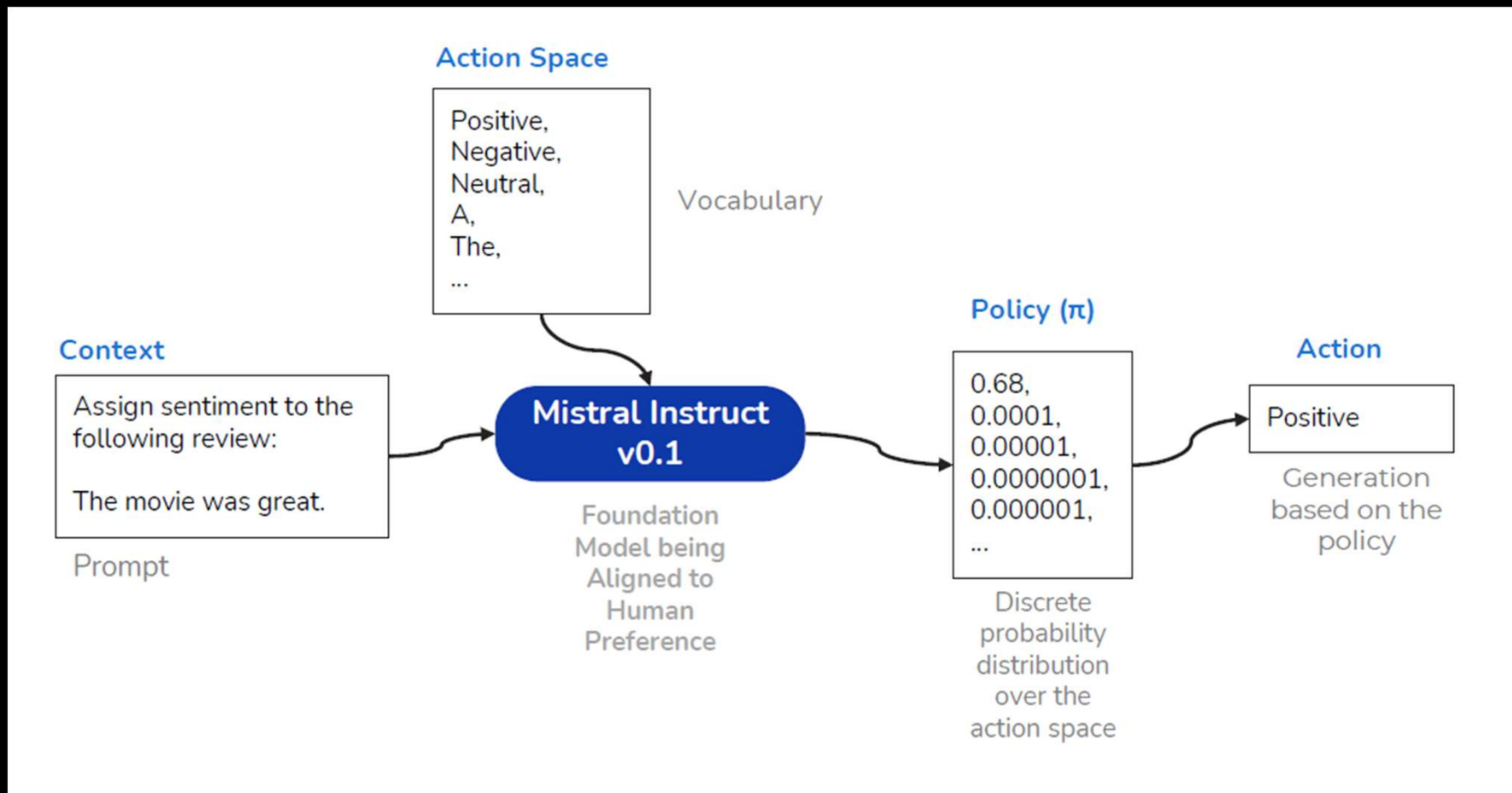
GRPO: Group Relative Policy Optimization

Reinforced Learning from Human Feedback (RLHF)

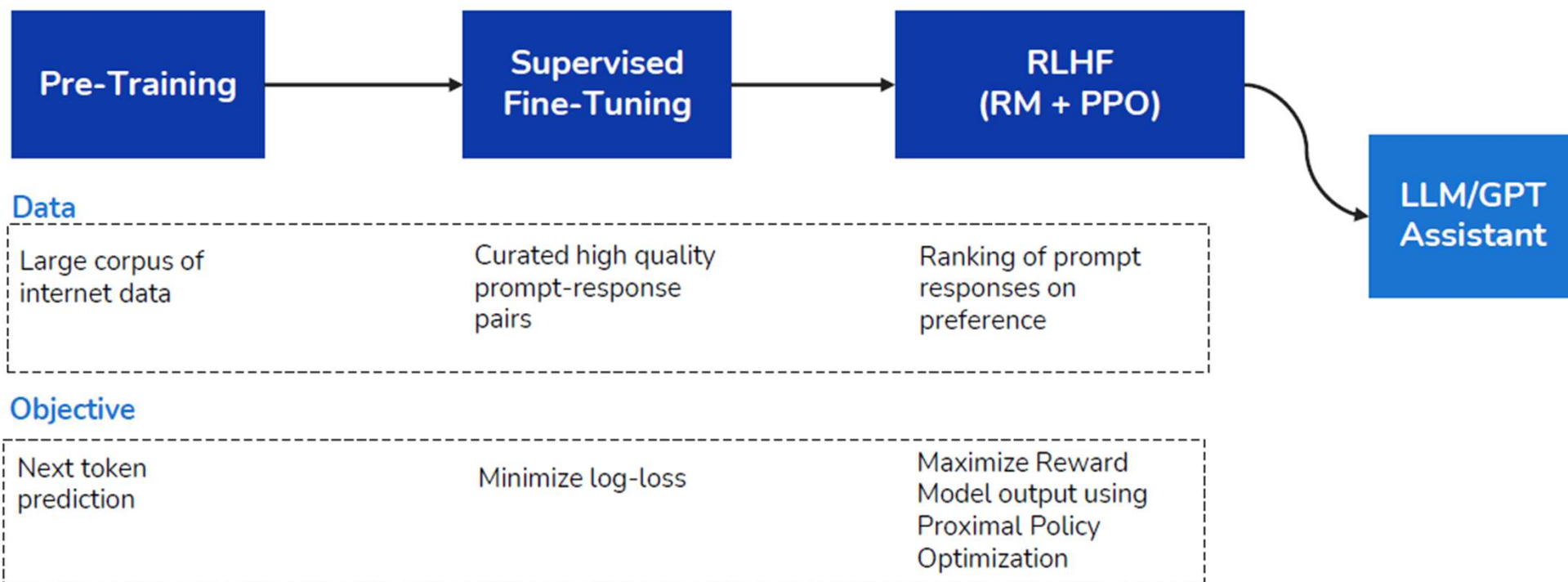
Reward modeling (RM). Starting from the SFT model with the final unembedding layer removed, we trained a model to take in a prompt and response, and output a scalar reward. In this paper we only use 6B RMs, as this saves a lot of compute, and we found that 175B RM training could be unstable and thus was less suitable to be used as the value function during RL (see Appendix C for more details).

Reinforcement learning (RL). Once again following Stiennon et al. (2020), we fine-tuned the SFT model on our environment using PPO (Schulman et al., 2017). The environment is a bandit environment which presents a random customer prompt and expects a response to the prompt. Given the prompt and response, it produces a reward determined by the reward model and ends the episode. In addition, we add a per-token KL penalty from the SFT model at each token to mitigate over-optimization of the reward model. The value function is initialized from the RM. We call these models “PPO.”

Alignment fine tuning techniques

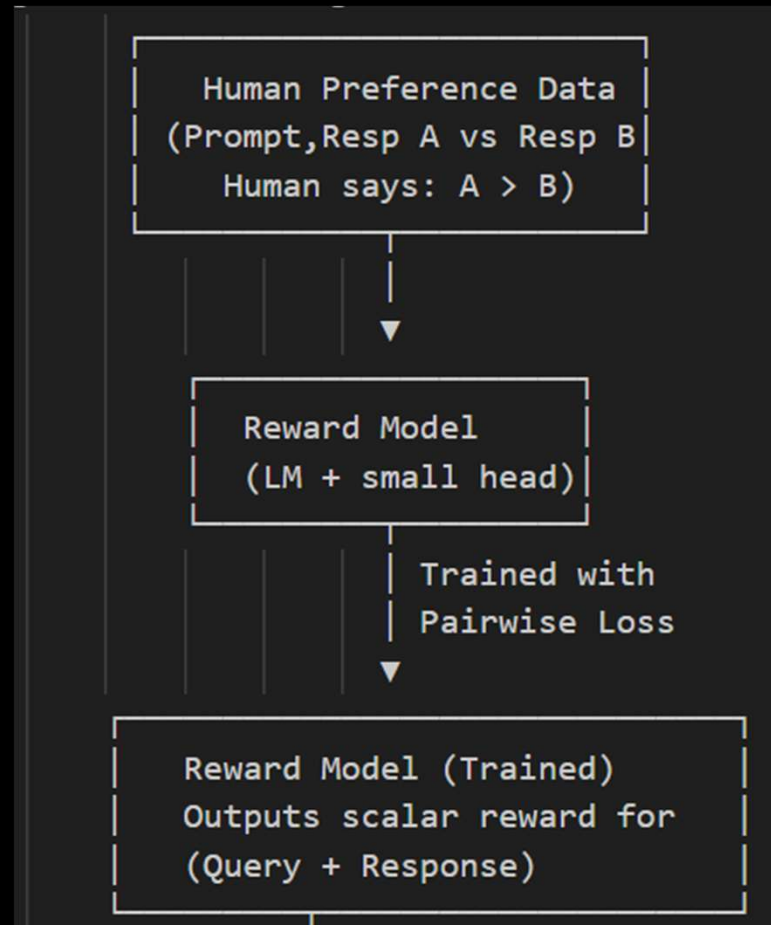


Alignment fine tuning techniques



Alignment fine tuning – Step 1: train the reward model

not needed for DPO



Alignment fine tuning Step 2: RLHF loop

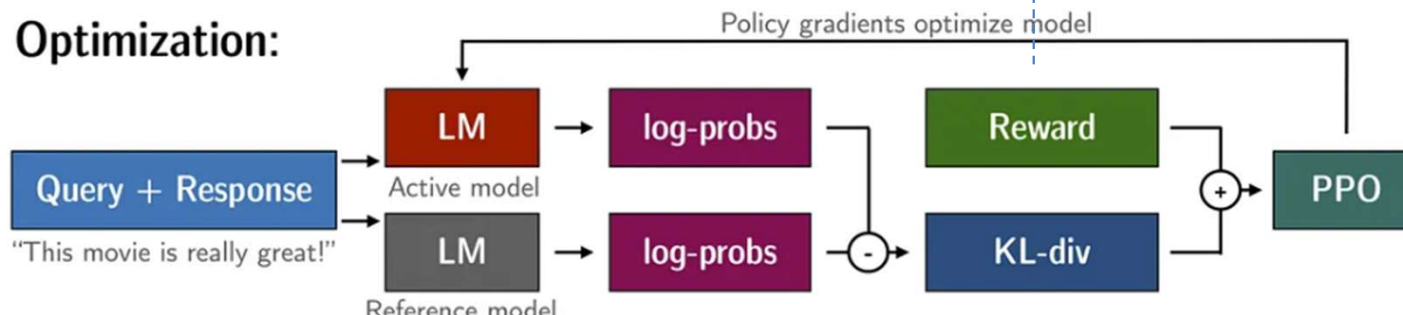
Rollout:



Evaluation:

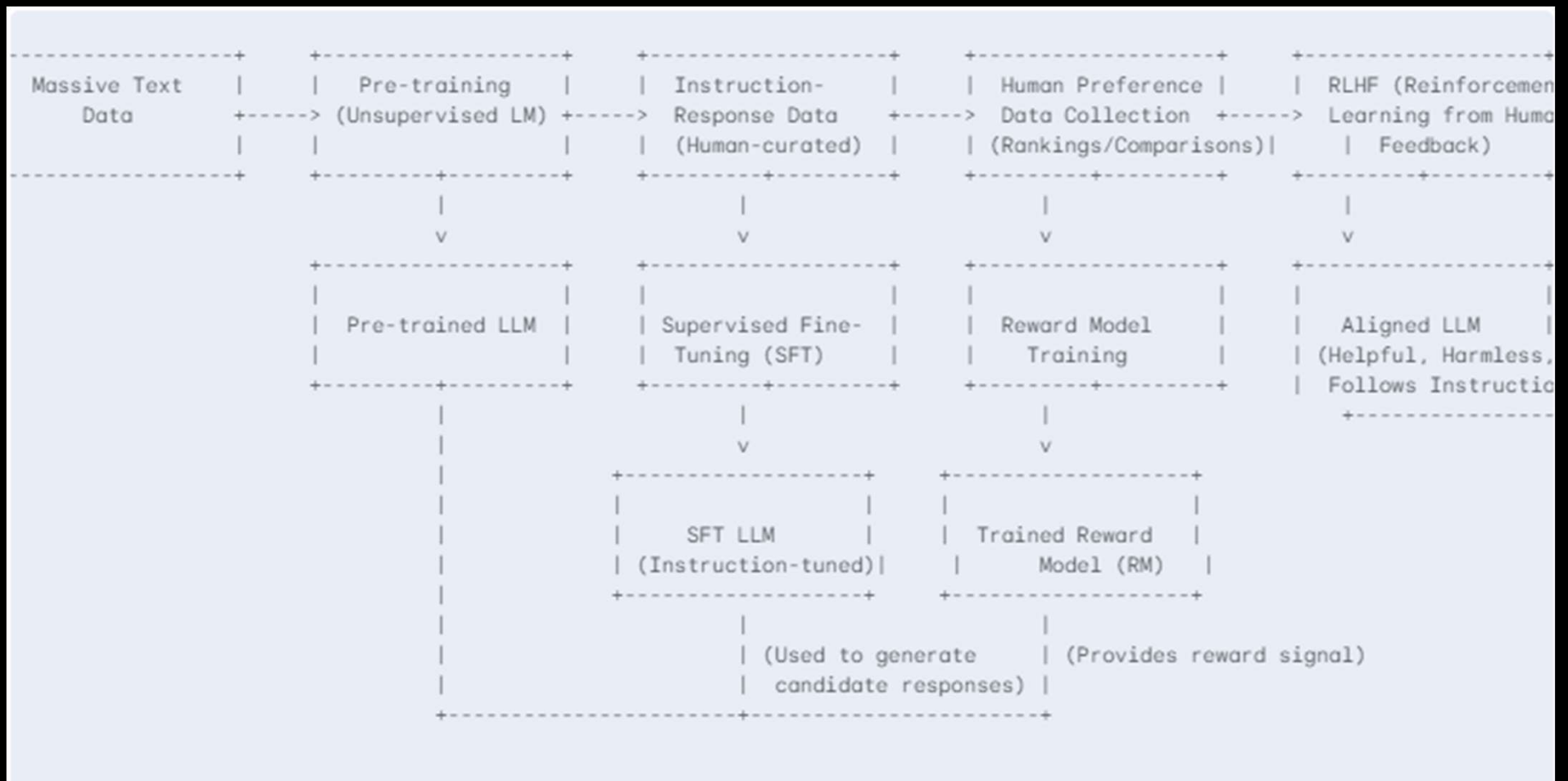


Optimization:



KL-div: Statistical method used to measure the difference between two probability distributions

Alignment fine tuning flow



Direct preference optimization (DPO)

```
dpo_dataset_dict = {  
  "prompt": ["hello", "how are you", ...],  
  "chosen": ["hi, nice to meet you", "I am fine", ...],  
  "rejected": ["leave me alone", "I am not fine", ...],  
}
```

Direct Preference Optimization (DPO)

x: "write me a poem about
the history of jazz"



preference data



maximum
likelihood



final LM

RLHF steps

1 Collect human feedback

A Reddit post is sampled from the Reddit TL;DR dataset.



Various policies are used to sample a set of summaries.



Two summaries are selected for evaluation.



A human judges which is a better summary of the post.



"j is better than k"

2 Train reward model

One post with two summaries judged by a human are fed to the reward model.



The reward model calculates a reward r for each summary.



r_j

r_k

The loss is calculated based on the rewards and human label, and is used to update the reward model.

$$\text{loss} = \log(\sigma(r_j - r_k))$$

"j is better than k"

3 Train policy with PPO

A new post is sampled from the dataset.



The policy π generates a summary for the post.



The reward model calculates a reward for the summary.



The reward is used to update the policy via PPO.

r

Steps for policy creation and optimization

Step 1

Collect demonstration data, and train a supervised policy.

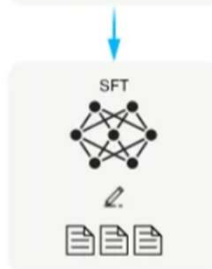
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



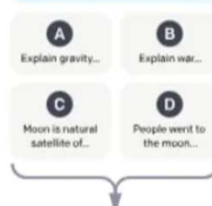
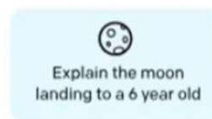
This data is used to fine-tune GPT-3 with supervised learning.



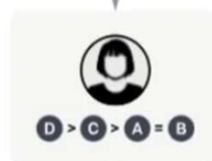
Step 2

Collect comparison data, and train a reward model.

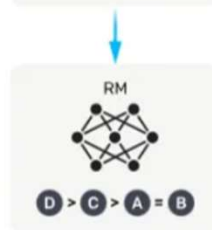
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



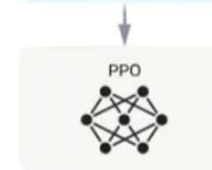
Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



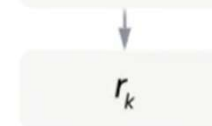
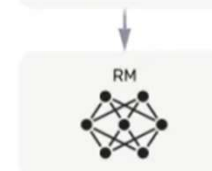
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



Reinforced learning from AI feedback (RLAIF)

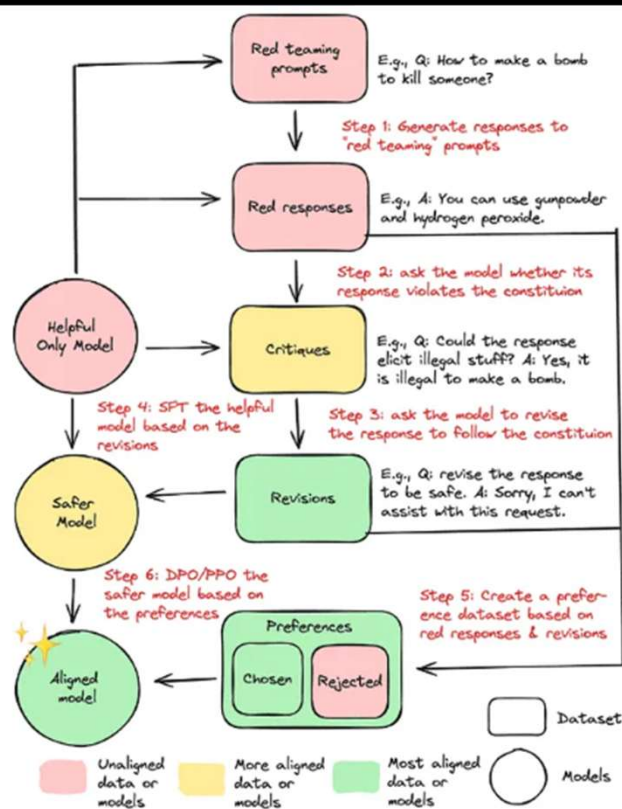


Figure 7. Reinforcement Learning from AI Feedback (RLAIF) and Constitutional AI (CAI)

Group Relative Policy Optimization (GRPO)

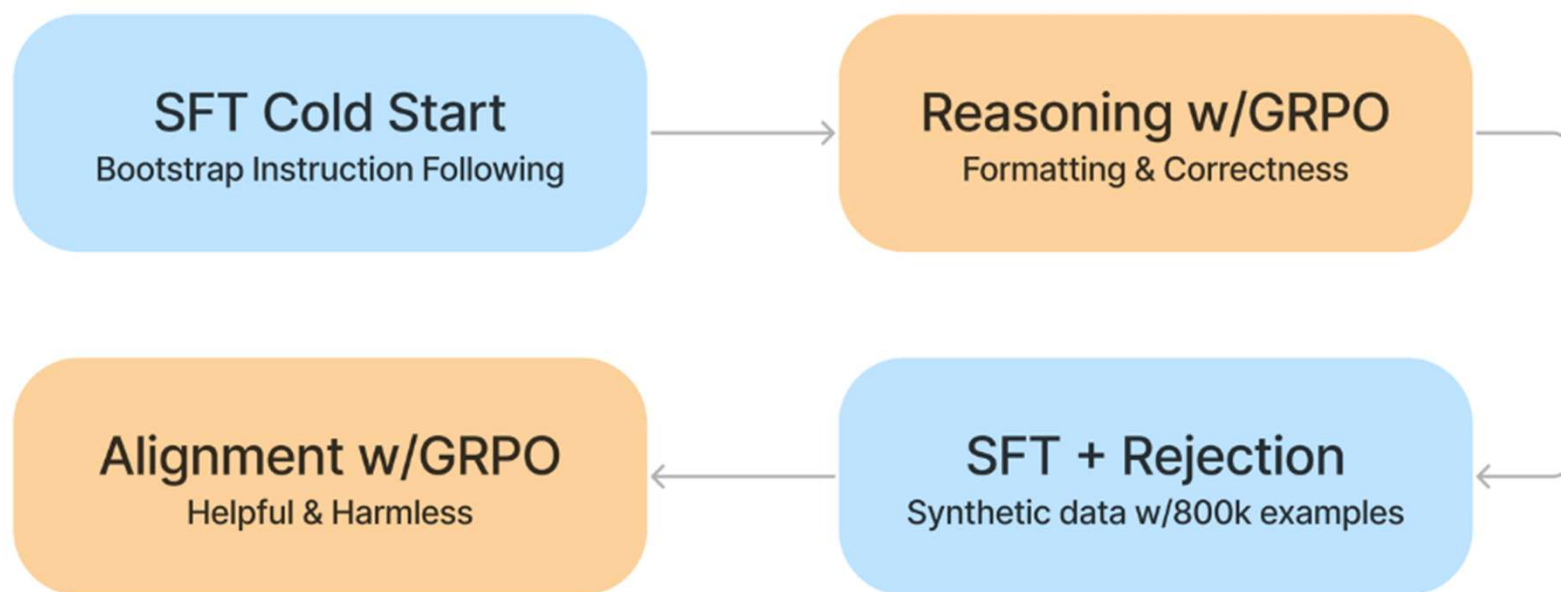
- Used by DeepSeek-R1
- At it's core aimed at improving reasoning ability
 - Train the model to have reasoning traces
<reasoning></reasoning>
 - Deterministic rewards for formatting, consistency, and correctness

<https://ghost.oxen.ai/why-grpo-is-important-and-how-it-works/>

Group Relative Policy Optimization (GRPO)



DeepSeek-R1 Training Pipeline



PPO Vs GRPO

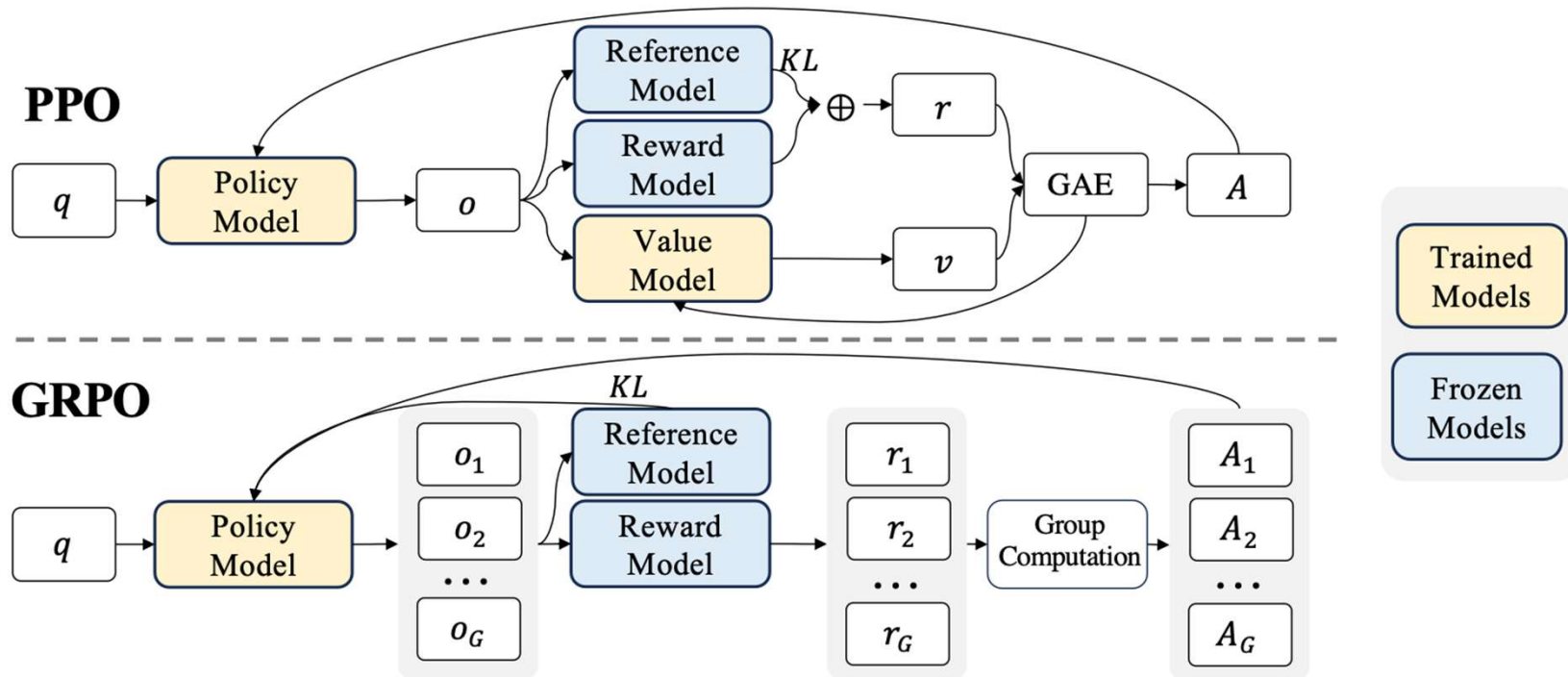
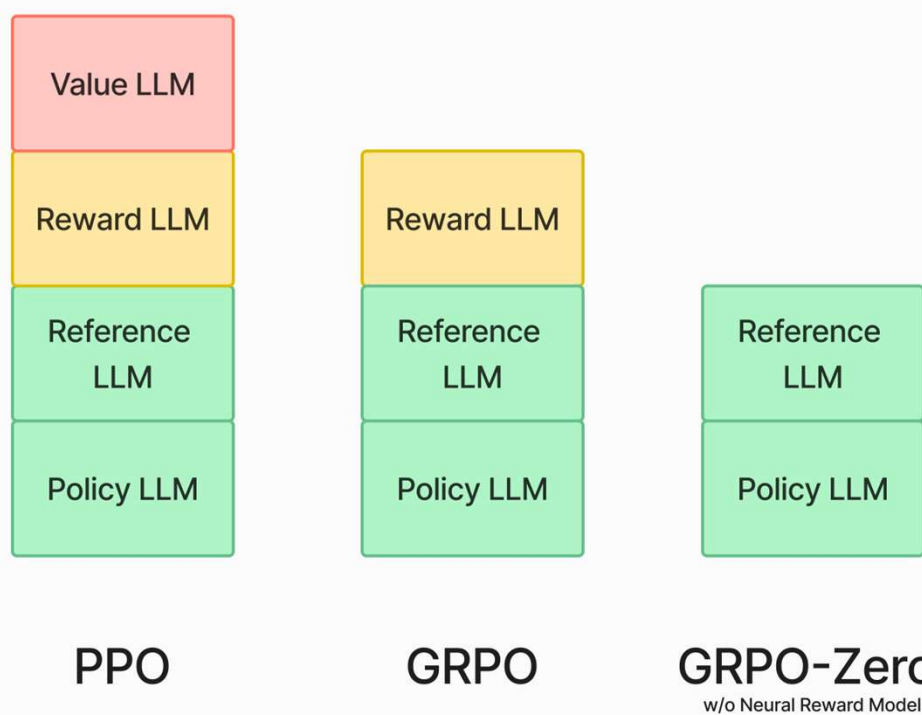


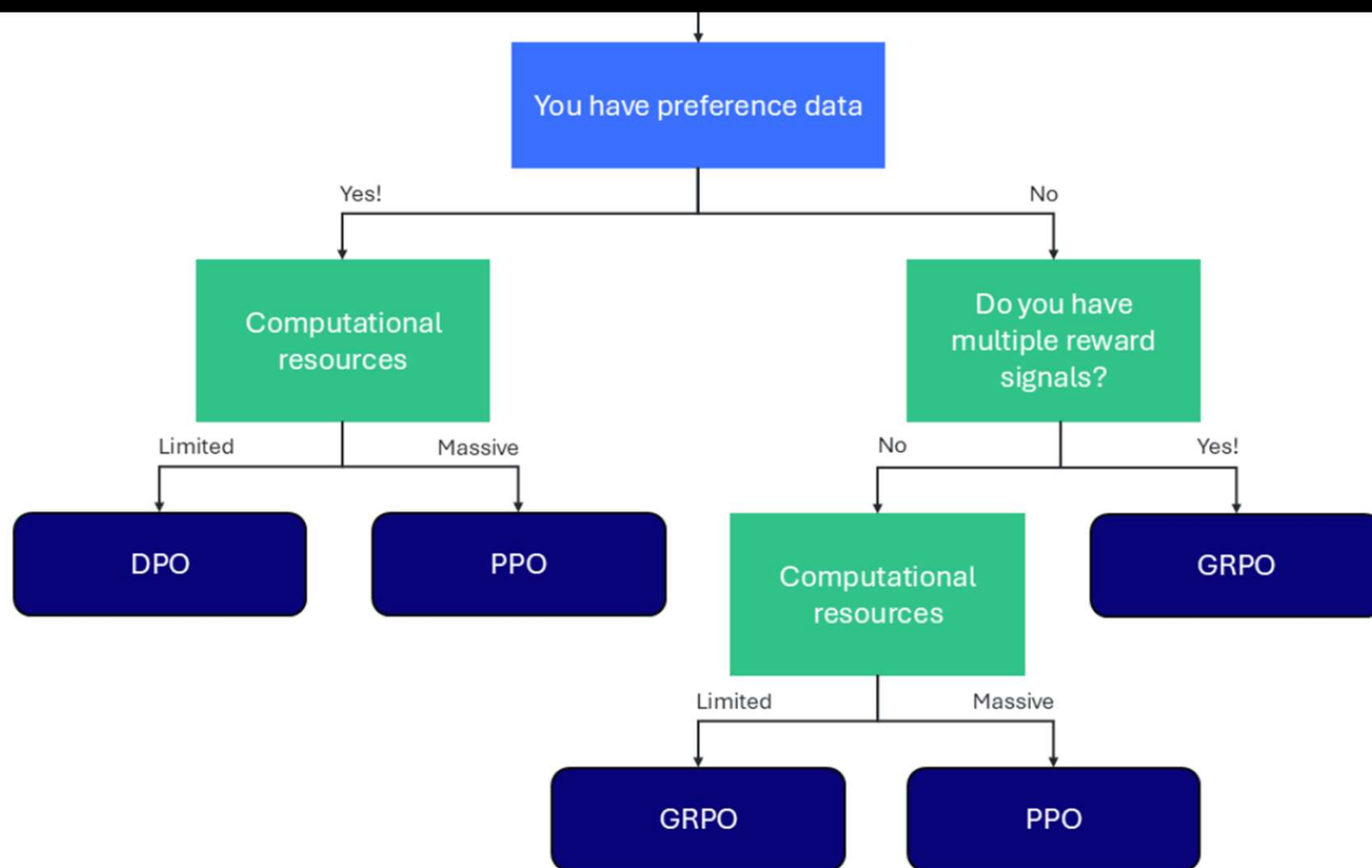
Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.

GRPO is compute and memory efficient

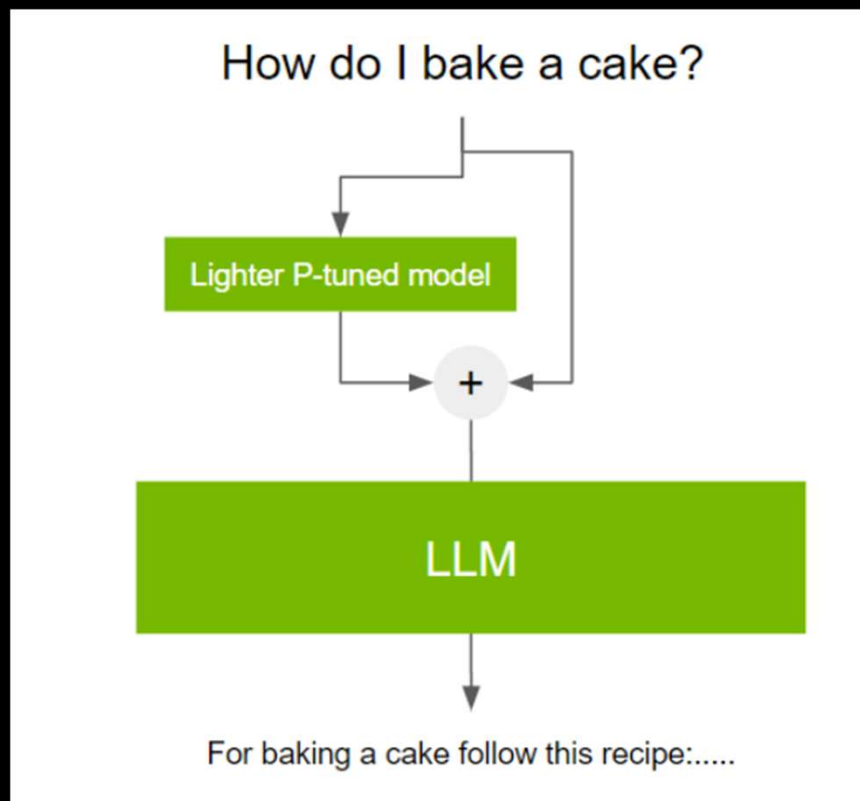
LLM Memory Usage (GPU VRAM)



Choosing the right RLHF technique



Prompt tuning: P-tuning, Prefix tuning



<https://ankitasinha0811.medium.com/ways-to-improve-an-llm-performance-while-keeping-the-llm-weights-frozen-7f67d740c5e2>

Fine tuning models with LoRA

<https://www.kaggle.com/code/nilaychauhan/fine-tune-gemma-models-in-keras-using-lora>

<https://dassum.medium.com/fine-tune-large-language-model-llm-on-a-custom-dataset-with-qlora-fb60abdeba07>